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Bearing block for a gear pump assembly

Abstract

A bearing block adapted for use in a gear pump assembly. The bearing block includes a bearing surface configured to face one or more gears. The bearing block further includes a bearing block body formed with a recess. The bearing block further includes an insert located within the recess. The insert includes an insert body including an outer surface. The insert body has a tapered shape that tapers outwardly from the bearing surface. The insert further includes a plurality of ridges disposed on the outer surface. The plurality of ridges defines one or more grooves therebetween. The recess includes a tapered shape complementary to the tapered shape of the insert body, such that the insert body engages with the bearing block body.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This specification is based upon and claims the benefit of priority from United Kingdom patent application number GB 2318901.2 filed on Dec. 12, 2023, the entire contents of which is incorporated herein by reference.

BACKGROUND

Technical Field

(2) The present disclosure relates to a bearing block, and in particular, to a bearing block adapted

for use in a gear pump assembly.

Description of the Related Art

(3) A fuel supply system of a gas turbine engine typically includes a gear pump assembly for pumping the fuel. The gear pump assembly generally operates over a large rotational speed operating range to provide fuel flow for various functions. The gear pump assembly includes a pair of intermeshing gears, one of which is arranged to drive the other. Rotation of the gears drive the fuel from an inlet side (i.e., low pressure side) of the gear pump assembly to an outlet side (i.e., high pressure side) of the gear pump assembly.

(4) Intermeshing teeth of the gears change the pressure of the fluid located therein. This may lead to formation of bubbles and subsequently cavitation erosion of the gear pump assembly. Each gear is supported by bearing blocks with a bearing surface. The bearing surfaces of the bearing blocks are commonly made of a lead bronze material and can be damaged by cavitation erosion.

(5) It is known to provide the bearing surfaces with inserts of a material of improved erosion resistance. One of the ways is to locate a peg in a corresponding opening or recess in the bearing block. The peg is disposed by an interference fit and held in position by a retaining pin. However, cavitation erosion of the surrounding bearing block may result in a leakage path being formed between the peg and the bearing block resulting in damage of the bearing block and in some cases, the teeth of the gears. This may lead to inefficient operation of the gear pump assembly.

(6) European patent EP1722103 B1 describes a method in which stepped or tapered inserts of a cavitation resistant material, such as aluminium bronze, is inserted into the bearing substrate (lead bronze) and is then encapsulated using a sprayed aluminium backing.

(7) The present disclosure describes an improvement to this method of retaining the insert for an improved bearing block for a gear pump assembly which is not susceptible to cavitation erosion.

SUMMARY

(8) According to a first aspect, a bearing block adapted for use in a gear pump assembly is disclosed. The bearing block includes a bearing surface configured to face one or more gears of the gear pump assembly. The bearing block further includes a bearing block body formed with a recess. The bearing block further includes an insert located within the recess of the bearing block body. The recess and insert are shaped so as to be of smaller dimensions at the bearing surface of the bearing block than at a point remote therefrom within the bearing block. The insert includes an insert body including an outer surface. The insert body has a tapered shape that tapers outwardly from the bearing surface. The insert further includes a plurality of ridges disposed on the outer surface of the insert body. The plurality of ridges defines one or more grooves therebetween. The recess includes a tapered shape complementary to the tapered shape of the insert body, such that the insert body engages with the bearing block body. A material of the bearing block body plastically deforms around the plurality of ridges and flows into each of the one or more grooves to improve a mechanical retention of the insert within the recess.

(9) As the material of the bearing block body plastically deforms around the plurality of ridges and flows into each of the one or more grooves upon engagement of the insert body with the bearing block body, there is minimal movement of the insert relative to the bearing block body. In other words, the plastic deformation of the material of the bearing block body around the plurality of ridges and the flow of the material of the bearing block body into each of the one or more grooves may mitigate any movement of the insert relative to the bearing block body. Therefore, as compared to conventional inserts for bearing blocks, the bearing block of the present disclosure includes the insert in such a way that the insert is firmly retained by the bearing block with minimal, if any, risk of the insert becoming loose.

(10) The improved mechanical retention of the insert in the bearing block body may avoid fluid leakage between high and low pressure sides of the gear pump assembly. This minimal fluid leakage between the high and low pressure sides may improve an efficiency of the gear pump assembly. By providing the bearing block of the present disclosure with improved mechanical

retention of the insert within the recess, any significant damage to the gear pump assembly may be prevented which could otherwise happen due to movement of the insert relative to the bearing block body.

(11) In some embodiments, the tapered shape of each of the insert body and the recess is frustoconical. Such tapered frustoconical shape of the each of the insert body and the recess may enhance mechanical retention of the insert within the recess.

(12) In some embodiments, each of the plurality of ridges is annular. The plurality of annular ridges defines corresponding one or more annular grooves. In case of the plurality of annular ridges, when the insert body engages with the bearing block body, the plastically deformed material of the bearing block body flows into each of the one or more annular grooves. The plastically deformed material may fully surround the insert body at the one or more annular grooves. Therefore, the presence of annular ridges as well as annular grooves may further improve the mechanical retention of the insert within the recess.

(13) In some embodiments, the insert is made of aluminium bronze. Aluminium bronze has a good cavitation erosion resistance whilst also having relatively good bearing properties.

(14) In some embodiments, the material of the bearing block body is lead bronze. As lead bronze is of low stiffness and strength, the bearing block body may plastically deform around the plurality of ridges upon engagement of the insert body with the bearing block body, thereby allowing improved mechanical retention of the insert within the recess. Lead bronze material also provides good bearing properties to the bearing block body.

(15) In some embodiments, the insert is flush or substantially flush with the bearing surface. As a result, there is minimum risk of causing undue wear or damage to the gears. As the insert lies flush with the bearing surface, formation of a leakage path between the high and low pressure sides of the gear pump assembly is avoided.

(16) According to a second aspect, a gear pump assembly for pumping a fluid is disclosed. The gear pump assembly includes an inlet configured to receive the fluid, an outlet, and a plurality of the bearing blocks of the first aspect. The gear pump assembly further includes a pair of intermeshing gears located between the plurality of bearing blocks and rotatable to pump the fluid from the inlet to the outlet. Providing the bearing blocks of the first aspect in the gear pump assembly may improve an efficiency and performance of the gear pump assembly.

(17) According to a third aspect, a fuel supply system of a gas turbine engine is disclosed. The fuel supply system includes the gear pump assembly of the second aspect for pumping a fuel. Providing the gear pump assembly of the second aspect in the fuel supply system of the gas turbine engine may improve an efficiency and performance of the fuel supply system.

(18) According to a fourth aspect, a gas turbine engine including the fuel supply system of the third aspect is disclosed. Providing the fuel supply system of the third aspect in the gas turbine engine may increase an efficiency of the gas turbine engine.

(19) According to a fifth aspect, a method of manufacturing the bearing block of the first aspect is disclosed. The method includes providing the recess in the bearing block body. The method further includes pushing the insert into the recess of the bearing block body. The method further includes subjecting the bearing block body to a hydrostatic pressure that causes the material of the bearing block body to plastically deform around the plurality of ridges and flow into each of the one or more grooves, thereby improving the mechanical retention of the insert within the recess.

(20) As the material of the bearing block body plastically deforms around the plurality of ridges and flows into each of the one or more grooves upon application of hydrostatic pressure on the bearing block body, there is minimal movement of the insert relative to the bearing block body in the bearing block manufactured by the method of the fifth aspect of the present disclosure. Therefore, as compared to conventional methods for manufacturing bearing blocks, the method of the present disclosure manufactures the bearing block in such a way that the insert is firmly retained by the bearing block with minimal, if any, risk of the insert becoming loose.

(21) As noted elsewhere herein, the present disclosure may relate to a gas turbine engine. Such a gas turbine engine may comprise an engine core comprising a turbine, a combustor, a compressor, and a core shaft connecting the turbine to the compressor. Such a gas turbine engine may comprise a fan (having fan blades) located upstream of the engine core.

(22) Arrangements of the present disclosure may be particularly, although not exclusively, beneficial for fans that are driven via a gearbox. Accordingly, the gas turbine engine may comprise a gearbox that receives an input from the core shaft and outputs drive to the fan so as to drive the fan at a lower rotational speed than the core shaft. The input to the gearbox may be directly from the core shaft, or indirectly from the core shaft, for example via a spur shaft and/or gear. The core shaft may rigidly connect the turbine and the compressor, such that the turbine and compressor rotate at the same speed (with the fan rotating at a lower speed). The gearbox may be a reduction gearbox (in that the output to the fan is a lower rotational rate than the input from the core shaft). Any type of gearbox may be used.

(23) The gas turbine engine as described and/or claimed herein may have any suitable general architecture. For example, the gas turbine engine may have any desired number of shafts that connect turbines and compressors, for example one, two or three shafts. Purely by way of example, the turbine connected to the core shaft may be a first turbine, the compressor connected to the core shaft may be a first compressor, and the core shaft may be a first core shaft. The engine core may further comprise a second turbine, a second compressor, and a second core shaft connecting the second turbine to the second compressor. The second turbine, second compressor, and second core shaft may be arranged to rotate at a higher rotational speed than the first core shaft.

(24) In such an arrangement, the second compressor may be positioned axially downstream of the first compressor. The second compressor may be arranged to receive (for example directly receive, for example via a generally annular duct) flow from the first compressor.

(25) In any gas turbine engine as described and/or claimed herein, a combustor may be provided axially downstream of the fan and compressor(s). For example, the combustor may be directly downstream of (for example at the exit of) the second compressor, where a second compressor is provided. By way of further example, the flow at the exit to the combustor may be provided to the inlet of the second turbine, where a second turbine is provided. The combustor may be provided upstream of the turbine(s).

(26) The or each compressor (for example the first compressor and second compressor as described above) may comprise any number of stages, for example multiple stages. Each stage may comprise a row of rotor blades and a row of stator vanes, which may be variable stator vanes (in that their angle of incidence may be variable). The row of rotor blades and the row of stator vanes may be axially offset from each other.

(27) The or each turbine (for example the first turbine and second turbine as described above) may comprise any number of stages, for example multiple stages. Each stage may comprise a row of rotor blades and a row of stator vanes. The row of rotor blades and the row of stator vanes may be axially offset from each other.

(28) Gas turbine engines in accordance with the present disclosure may have any desired bypass ratio, where the bypass ratio is defined as the ratio of the mass flow rate of the flow through the bypass duct to the mass flow rate of the flow through the core at cruise conditions. The bypass duct may be substantially annular. The bypass duct may be radially outside the engine core. The radially outer surface of the bypass duct may be defined by a nacelle and/or a fan case.

(29) Specific thrust of an engine may be defined as the net thrust of the engine divided by the total mass flow through the engine. At cruise conditions, the specific thrust of an engine described and/or claimed herein may be less than (or in the order of) any of the following: 110 Nkg-1s, 105 Nkg-1s, 100 Nkg-1s, 95 Nkg-1s, 90 Nkg-1s, 85 Nkg-1s or 80 Nkg-1s. The specific thrust may be in an inclusive range bounded by any two of the values in the previous sentence (i.e., the values may form upper or lower bounds), for example in the range of from 80 Nkg-1s to 100 Nkg-1s, or 85

Nkg-1s to 95 Nkg-1s. Such engines may be particularly efficient in comparison with conventional gas turbine engines.

(30) A fan blade and/or aerofoil portion of a fan blade described and/or claimed herein may be manufactured from any suitable material or combination of materials. For example, at least a part of the fan blade and/or aerofoil may be manufactured at least in part from a composite, for example a metal matrix composite and/or an organic matrix composite, such as carbon fibre.

(31) The fan of a gas turbine as described and/or claimed herein may have any desired number of fan blades, for example 14, 16, 18, 20, 22, 24 or 26 fan blades.

(32) The skilled person will appreciate that except where mutually exclusive, a feature or parameter described in relation to any one of the above aspects may be applied to any other aspect.

Furthermore, except where mutually exclusive, any feature or parameter described herein may be applied to any aspect and/or combined with any other feature or parameter described herein.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Embodiments will now be described by way of example only, with reference to the Figures, in which:

(2) FIG. 1 is a sectional side view of a gas turbine engine;

(3) FIG. 2 is a sectional view of a gear pump assembly for pumping a fluid in the gas turbine engine of FIG. 1, according to an embodiment of the present disclosure;

(4) FIG. 3 is a top view of a bearing block of the gear pump assembly;

(5) FIG. 4 is a sectional side view of the bearing block of FIG. 3;

(6) FIG. 5 is a side view of an insert of the bearing block of FIG. 3;

(7) FIG. 6 is a side view of an insert, according to another embodiment of the present disclosure; and

(8) FIG. 7 is a flowchart for a method for manufacturing the bearing block of FIG. 3, according to an embodiment of the present disclosure.

DETAILED DESCRIPTION

(9) Aspects and embodiments of the present disclosure will now be discussed with reference to the accompanying Figures. Further aspects and embodiments will be apparent to those skilled in the art.

(10) FIG. 1 illustrates a gas turbine engine **10** having a principal rotational axis **9**. The engine **10** comprises an air intake **12** and a propulsive fan **23** that generates two airflows: a core airflow A and a bypass airflow B. The gas turbine engine **10** comprises an engine core **11** that receives the core airflow A. The engine core **11** comprises, in axial flow series, a low pressure compressor **14**, a high pressure compressor **15**, a combustion equipment **16**, a high pressure turbine **17**, a low pressure turbine **19**, and a core exhaust nozzle **20**. A nacelle **21** surrounds the gas turbine engine **10** and defines a bypass duct **22** and a bypass exhaust nozzle **18**. The bypass airflow B flows through the bypass duct **22**. The fan **23** is attached to and driven by the low pressure turbine **19** via a shaft **26** and an epicyclic gearbox **30**.

(11) In use, the core airflow A is accelerated and compressed by the low pressure compressor **14** and directed into the high pressure compressor **15** where further compression takes place. The compressed air exhausted from the high pressure compressor **15** is directed into the combustion equipment **16** where it is mixed with fuel and the mixture is combusted. The resultant hot combustion products then expand through, and thereby drive, the high pressure and low pressure turbines **17**, **19** before being exhausted through the core exhaust nozzle **20** to provide some propulsive thrust. The high pressure turbine **17** drives the high pressure compressor **15** by a suitable interconnecting shaft **27**. The fan **23** generally provides the majority of the propulsive thrust. The

epicyclic gearbox **30** is a reduction gearbox.

(12) Note that the terms “low pressure turbine” and “low pressure compressor” as used herein may be taken to mean the lowest pressure turbine stages and lowest pressure compressor stages (i.e., not including the fan **23**) respectively and/or the turbine and compressor stages that are connected together by the interconnecting shaft **26** with the lowest rotational speed in the engine **10** (i.e., not including the gearbox output shaft that drives the fan **23**). In some literature, the “low pressure turbine” and “low pressure compressor” referred to herein may alternatively be known as the “intermediate pressure turbine” and “intermediate pressure compressor”. Where such alternative nomenclature is used, the fan **23** may be referred to as a first, or lowest pressure, compression stage.

(13) Other gas turbine engines to which the present disclosure may be applied may have alternative configurations. For example, such engines may have an alternative number of compressors and/or turbines and/or an alternative number of interconnecting shafts. By way of further example, the gas turbine engine shown in FIG. **1** has a split flow nozzle **18, 20** meaning that the flow through the bypass duct **22** has its own nozzle **18** that is separate to and radially outside the core exhaust nozzle **20**. However, this is not limiting, and any aspect of the present disclosure may also apply to engines in which the flow through the bypass duct **22** and the flow through the core **11** are mixed, or combined, before (or upstream of) a single nozzle, which may be referred to as a mixed flow nozzle. One or both nozzles (whether mixed or split flow) may have a fixed or variable area. Whilst the described example relates to a turbofan engine, the disclosure may apply, for example, to any type of gas turbine engine, such as an open rotor (in which the fan stage is not surrounded by a nacelle) or turboprop engine, for example. In some arrangements, the gas turbine engine **10** may not comprise a gearbox **30**.

(14) The geometry of the gas turbine engine **10**, and components thereof, is defined by a conventional axis system, comprising an axial direction (which is aligned with the rotational axis **9**), a radial direction (in the bottom-to-top direction in FIG. **1**), and a circumferential direction (perpendicular to the page in the FIG. **1** view). The axial, radial, and circumferential directions are mutually perpendicular.

(15) In addition, the present disclosure is equally applicable to aero gas turbine engines, marine gas turbine engines, and land-based gas turbine engines.

(16) The gas turbine engine **10** further includes a fuel supply system **50** (shown schematically in FIG. **1**) to supply adequate amount of fuel to the gas turbine engine **10**. The fuel supply system **50** includes a gear pump assembly **100** for pumping a fuel or a fluid. The fuel supply system **50** may also include other parts (not shown), such as an auxiliary gear box attached to the engine core **11**.

(17) FIG. **2** shows a sectional view of the gear pump assembly **100**, according to an embodiment of the present disclosure. The gear pump assembly **100** may be a twin pinion gear fuel pump. Here, the gear pump assembly **100** is shown for use in the fuel supply system **50** of the gas turbine engine **10**. However, in other embodiments, the gear pump assembly **100** can be used in other applications as well.

(18) The gear pump assembly **100** includes a housing **101**, an inlet **102** configured to receive the fluid, and an outlet **104** to discharge the fluid. The gear pump assembly **100** further includes a pair of intermeshing gears **106, 108** disposed in the housing **101**. The gears **106, 108** are rotatable to pump the fluid from the inlet **102** to the outlet **104**. The gear pump assembly **100** further includes a plurality of bearing blocks **110** adapted for use in the gear pump assembly **100**. The plurality of bearing blocks **110** support the gears **106, 108**. The gears **106, 108** are located between the plurality of bearing blocks **110**.

(19) Each of the gears **106, 108** is provided with corresponding teeth **112, 114** arranged to mesh with one another. The gear **106** is mounted upon a drive shaft **116** which is driven for rotation to cause rotation of the gear **106**. The other gear **108** is also mounted for rotation within the housing **101**. The rotation of the gear **106** causes rotation of the gear **108** by virtue of the intermeshing of

the teeth **112**, **114**. In use, fluid is driven by the gears **106**, **108** from the low pressure side (inlet side) of the gear pump assembly **100** between the gears **106**, **108** and the housing **101** to the high pressure side (outlet side) thereof. Some gears, bearings, and shafts are not shown in FIG. 2 for illustrative purposes.

(20) FIG. 3 is a top view of the bearing block **110**, according to an embodiment of the present disclosure. FIG. 4 is a sectional side view of the bearing block **110**, according to an embodiment of the present disclosure. Referring to FIGS. 2 to 4, the bearing block **110** includes a bearing surface **118** (shown in FIG. 4) configured to face one or more gears (i.e., the gears **106**, **108**) of the gear pump assembly **100**.

(21) Generally, the intermeshed teeth **112**, **114** and bearing surface **118** of the bearing block **110** together serve to define isolated chambers (not shown) which move, upon rotation of the gears **106**, **108** and cause variations in the fluid pressure therein. The variations in fluid pressure can give rise to formation of bubbles in the fluid in the chambers, and the subsequent collapse of the bubbles. The bearing block **110** is of such design and material that there is negligible cavitation erosion of the bearing block **110**.

(22) The bearing block **110** further includes a bearing block body **120** formed with a recess **122**. The bearing block body **120** may be a cylindrical body. In some embodiments, the material of the bearing block body **120** is lead bronze. Lead bronze material also provides good bearing properties to the bearing block body **120**. In some other embodiments, the material of the bearing block body **120** may be aluminium bronze.

(23) The bearing block **110** further includes an insert **124** located within the recess **122** of the bearing block body **120**. FIG. 5 is a side view of the insert **124**, according to an embodiment of the present disclosure. Referring to FIGS. 2 to 5, the insert **124** is made of a material better capable of withstanding cavitation erosion than the lead bronze material of the bearing block body **120**. In some embodiments, the insert **124** is made of aluminium bronze. Aluminium bronze has a good cavitation erosion resistance whilst also having relatively good bearing properties.

(24) In some embodiments, the insert **124** is flush or substantially flush with the bearing surface **118**. As a result, there is minimum risk of causing undue wear or damage to the gears **106**, **108**. As the insert **124** lies flush with the bearing surface **118**, formation of a leakage path between the high and low pressure sides of the gear pump assembly **100** is avoided. In some cases, a machining operation may be carried out to result in the insert **124** and the bearing surface **118** lying flush with one another. The insert **124** defines a first end **124a** disposed adjacent to the bearing surface **118** and a second end **124b** disposed within the bearing block **110** and remote from the bearing surface **118**.

(25) The recess **122** and insert **124** are shaped so as to be of smaller dimensions at the bearing surface **118** of the bearing block **110** than at a point remote therefrom within the bearing block **110**. The dimensions of the insert **124** and the recess **122** are such that, when the insert **124** has been fully introduced into the recess **122**, part of the insert **124** lies flush with the bearing surface **118**.

(26) The insert **124** includes an insert body **126** including an outer surface **128**. The insert body **126** has a tapered shape that tapers outwardly from the bearing surface **118**. In some embodiments, the tapered shape of each of the insert body **126** and the recess **122** is frustoconical. Such tapered frustoconical shape of each of the insert body **126** and the recess **122** may enhance mechanical retention of the insert **124** within the recess **122**. In some embodiments, the insert **124** and the recess **122** may be of double dove-tail shaped form as viewed from the bearing surface **118**.

(27) The insert **124** further includes a plurality of ridges **130** disposed on the outer surface **128** of the insert body **126**. The plurality of ridges **130** is disposed proximal to the first end **124a** and distal to the second end **124b** of the insert **124**. The plurality of ridges **130** defines one or more grooves **132** therebetween. In other words, every two adjacent ridges **130** from the plurality of ridges **130** define one groove **132** therebetween. In the illustrated embodiment of FIGS. 4 and 5, the insert **124** includes three ridges **130** defining two grooves **132** therebetween. However, a number of the ridges

130 may be varied as per application requirements. In some embodiments, the plurality of ridges **130** extends along a portion of the length between the first end **124a** and the second end **124b** of the insert **124**. In some embodiments, the plurality of ridges **130** extends along 30% of the length between the first end **124a** and the second end **124b** of the insert **124**.

(28) The recess **122** includes a tapered shape complementary to the tapered shape of the insert body **126**, such that the insert body **126** engages with the bearing block body **120**. The material of the bearing block body **120** plastically deforms around the plurality of ridges **130** and flows into each of the one or more grooves **132** to improve a mechanical retention of the insert **124** within the recess **122**. As the material (e.g., lead bronze) of the bearing block body **120** is of low stiffness and strength, the bearing block body **120** may plastically deform around the plurality of ridges **130** upon engagement of the insert body **126** with the bearing block body **120**, thereby allowing improved mechanical retention of the insert **124** within the recess **122**.

(29) As the material of the bearing block body **120** plastically deforms around the plurality of ridges **130** and flows into each of the one or more grooves **132** upon engagement of the insert body **126** with the bearing block body **120**, there is minimal movement of the insert **124** relative to the bearing block body **120**. In other words, the plastic deformation of the material of the bearing block body **120** around the plurality of ridges **130** and the flow of the material of the bearing block body **120** into each of the one or more grooves **132** may mitigate any movement of the insert **124** relative to the bearing block body **120**. Therefore, as compared to conventional inserts for bearing blocks, the bearing block **110** of the present disclosure includes the insert **124** in such a way that the insert **124** is firmly retained by the bearing block **110** with minimal, if any, risk of the insert **124** becoming loose.

(30) The improved mechanical retention of the insert **124** in the bearing block body **120** may avoid fluid leakage between high and low pressure sides of the gear pump assembly **100**. This minimal fluid leakage between the high and low pressure sides may improve an efficiency of the gear pump assembly **100**. By providing the bearing block **110** of the present disclosure with improved mechanical retention of the insert **124** within the recess **122**, any significant damage to the gear pump assembly **100** may be prevented which could otherwise happen due to movement of the insert **124** relative to the bearing block body **120**. In some embodiments, each of the plurality of ridges **130** is annular. The plurality of annular ridges **130** defines corresponding one or more annular grooves **132**. In case of the plurality of annular ridges **130**, when the insert body **126** engages with the bearing block body **120**, the plastically deformed material of the bearing block body **120** flows into each of the one or more annular grooves **132**. The plastically deformed material may fully surround the insert body **126** at the one or more annular grooves **132**. Therefore, the presence of annular ridges **130** as well as annular grooves **132** may further improve the mechanical retention of the insert **124** within the recess **122**.

(31) FIG. 6 is a side view of an insert **124'**, according to another embodiment of the present disclosure. The insert **124'** is substantially similar to the insert **124** of FIG. 5, with common components being referred to by the same numerals. However, in the insert **124'**, the one or more grooves **132** are provided by machining a portion of the outer surface **128** (i.e., outer diameter in this embodiment) of the insert body **126**. In this embodiment, the one or more grooves **132** define the plurality of ridges **130** therebetween. In other words, it can again be stated that the plurality of ridges **13** define the one or more grooves **132** therebetween.

(32) FIG. 7 is a flowchart for a method **200** for manufacturing the bearing block **110** of FIG. 3, according to an embodiment of the present disclosure. Referring to FIGS. 2 to 7, at step **202**, the method **200** includes providing the recess **122** in the bearing block body **120**. The recess **122** may be provided by machining the bearing block body **120**. At step **204**, the method **200** includes pushing the insert **124** (or the insert **124'** of FIG. 6) into the recess **122** of the bearing block body **120**.

(33) At step **206**, the method **200** further includes subjecting the bearing block body **120** to a

hydrostatic pressure that causes the material of the bearing block body **120** to plastically deform around the plurality of ridges **130** and flow into each of the one or more grooves **132**, thereby improving the mechanical retention of the insert **124** (or the insert **124'**) within the recess **122**.

(34) As the material of the bearing block body **120** plastically deforms around the plurality of ridges **130** and flows into each of the one or more grooves **132** upon application of hydrostatic pressure on the bearing block body **120**, there is minimal movement of the insert **124** relative to the bearing block body **120** in the bearing block **110** manufactured by the method **200** of the present disclosure. Therefore, as compared to conventional methods for manufacturing bearing blocks, the method **200** manufactures the bearing block **110** in such a way that the insert **124** is firmly retained by the bearing block **110** with minimal if any risk of the insert becoming loose.

(35) It will be understood that the invention is not limited to the embodiments above described and various modifications and improvements can be made without departing from the concepts described herein. Except where mutually exclusive, any of the features may be employed separately or in combination with any other features and the disclosure extends to and includes all combinations and sub-combinations of one or more features described herein.

Claims

1. A bearing block adapted for use in a gear pump assembly, the bearing block comprising: a bearing surface configured to face one or more gears of the gear pump assembly; a bearing block body formed with a recess, and an insert located within the recess of the bearing block body, the recess and insert being shaped so as to be of smaller dimensions at the bearing surface of the bearing block than at a point remote therefrom within the bearing block, wherein the insert comprises: an insert body comprising an outer surface, the insert body having a tapered shape that tapers outwardly from the bearing surface; and a plurality of ridges disposed on the outer surface of the insert body, the plurality of ridges defining one or more grooves therebetween; wherein the recess comprises a tapered shape complementary to the tapered shape of the insert body, such that the insert body engages with the bearing block body, and wherein a material of the bearing block body plastically deforms around the plurality of ridges and flows into each of the one or more grooves to improve a mechanical retention of the insert within the recess.
 2. The bearing block of claim 1, wherein the tapered shape of each of the insert body and the recess is frustoconical.
 3. The bearing block of claim 1, wherein each of the plurality of ridges is annular.
 4. The bearing block of claim 1, wherein the insert is made of aluminium bronze.
 5. The bearing block of claim 1, wherein the material of the bearing block body is lead bronze.
 6. The bearing block of claim 1, wherein the insert is flush or substantially flush with the bearing surface.
 7. A gear pump assembly for pumping a fluid, the gear pump assembly comprising: an inlet configured to receive the fluid; an outlet; a plurality of the bearing blocks of claim 1; and a pair of intermeshing gears located between the plurality of bearing blocks and rotatable to pump the fluid from the inlet to the outlet.
 8. A fuel supply system of a gas turbine engine, the fuel supply system comprising the gear pump assembly of claim 7 for pumping a fuel.
 9. A method of manufacturing the bearing block of claim 1, the method comprising the steps of: providing the recess in the bearing block body; pushing the insert into the recess of the bearing block body; and subjecting the bearing block body to a hydrostatic pressure that causes the material of the bearing block body to plastically deform around the plurality of ridges and flow into each of the one or more grooves, thereby improving the mechanical retention of the insert within the recess.
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